HOCHSCHULE
RAVENSBURG-WEINGARTEN
UNIVERSITY
OF APPLIED SCIENCES

Electronic Circuit Design

Slides to the lecture

Chapter 5: Bridge Circuits

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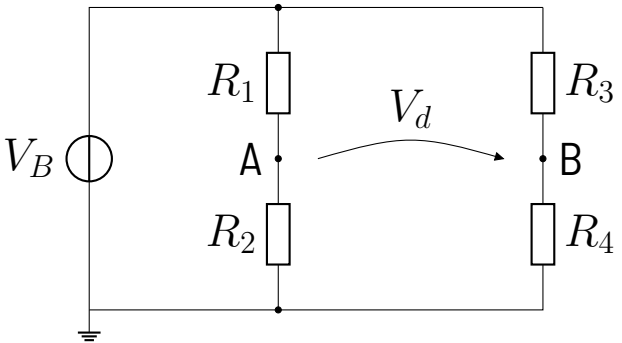
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5 Bridge Circuits

5.1 The Wheatstone Bridge

Two voltage dividers



Bridge output

$$V_d = V_A - V_B = V_B \left(\frac{R_2}{R_1 + R_2} - \frac{R_4}{R_3 + R_4} \right)$$

Balance condition

- the bridge is **balanced** if $V_d = 0$:

Balance condition

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} \quad \Leftrightarrow \quad R_1 R_4 = R_2 R_3$$

- the balance does **not** depend on V_B: supply noise and drift do not appear at the output
- null method: adjust a known resistor until $V_d = 0$, then $R_2 = R_1 R_4 / R_3$
- deflection method: fixed resistors, balanced at the reference point (e.g. $0\,°\text{C}$), measure V_d

5.2 Output Voltage and Linearity

Quarter bridge with equal resistors

- $R_1 = R_3 = R_4 = R_0$, sensor $R_2 = R_0(1 + x)$:

$$V_d = V_B \left(\frac{1+x}{2+x} - \frac{1}{2} \right) = \frac{V_B}{4} \cdot \frac{x}{1 + x/2}$$

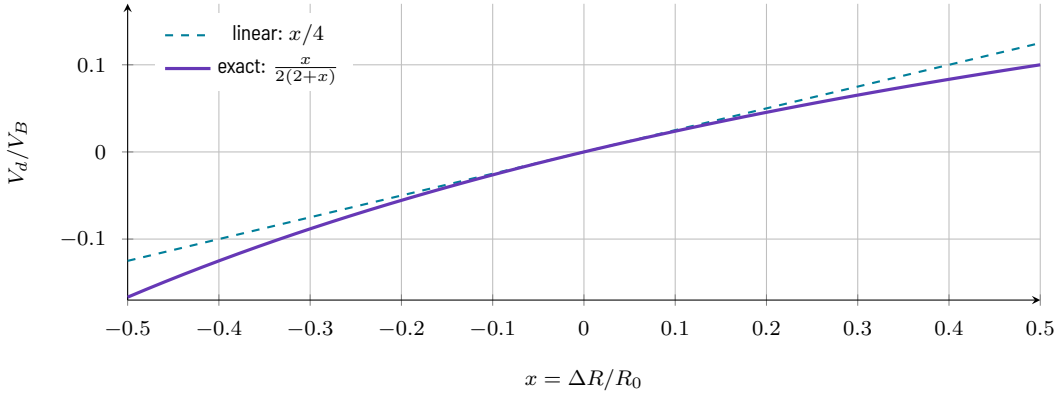
Quarter bridge, small signal

$$V_d \approx \frac{V_B}{4} \frac{\Delta R}{R_0} \qquad \text{relative error} \approx -\tfrac{x}{2}$$

- example: $V_B = 5\,\text{V}$, $x = 1\,\%$: $V_d = 12.44\,\text{mV}$ (estimate $12.5\,\text{mV}$)
- $x = 50\,\%$: $V_d = 0.500\,\text{V}$ instead of $0.625\,\text{V}$

▶ Falstad: `examples/bridges/wheatstone.txt`

The bridge is not linear



- positive x: output too small, negative x: too large
- strain gauges ($x < 0.5\,\%$): negligible; Pt100 over 100 K ($x = 38.5\,\%$): $16\,\%$

Unequal arms: trade sensitivity for linearity

- $R_1 = R_3 = m\,R_0$, $R_4 = R_0$, sensor $R_0(1+x)$:

$$V_d = V_B \frac{m\,x}{(m+1)(m+1+x)} \approx V_B \frac{m}{(m+1)^2}\,x$$

- sensitivity $m/(m+1)^2$: maximum $1/4$ at $m = 1$; $m = 39$: 0.024
- relative nonlinearity $\approx -x/(m+1)$: $m = 39$ makes it 20 times smaller
- large m also limits the sensor current: $I = V_B/(R_1 + R_{\text{sensor}})$ (self-heating!)

Exercise 5.1: Wheatstone bridge

A bridge with $R_1 = R_3 = R_4 = 1\,\text{k}\Omega$ is supplied with $V_B = 5\,\text{V}$. The resistor $R_2 = R_x$ is the sensor.

a) Calculate V_d exactly and with the small-signal formula for $R_x = 1010\,\Omega$.
b) Repeat for $R_x = 1500\,\Omega$. How large is the error of the formula?
c) Check both values in Falstad.

Solution 5.1

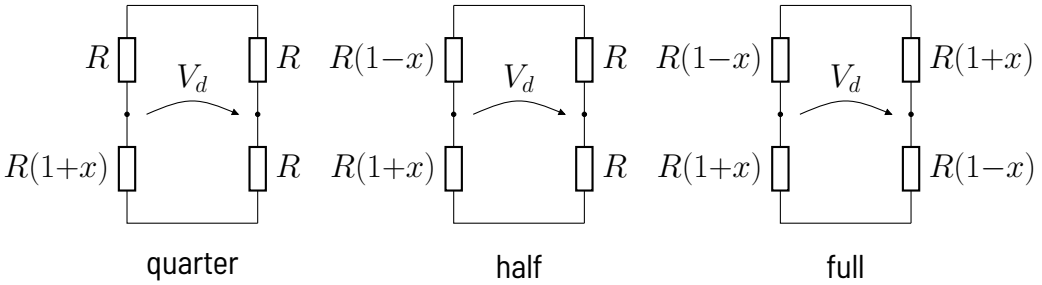
a) $V_d = 5\,(1010/2010 - 0.5) = 12.44\,\text{mV}$; formula: $\frac{5}{4} \cdot 0.01 = 12.5\,\text{mV}$ ($0.5\,\%$ too high, $\approx x/2$).

b) $V_d = 5\,(1500/2500 - 0.5) = 0.500\,\text{V}$; formula: $\frac{5}{4} \cdot 0.5 = 0.625\,\text{V}$: $25\,\%$ too high.

c) ▶ Falstad: `examples/bridges/wheatstone.txt`: the voltmeter shows $12.44\,\text{mV}$; with the slider at $1500\,\Omega$ it shows $0.500\,\text{V}$.

5.3 Half and Full Bridges: Strain Gauges

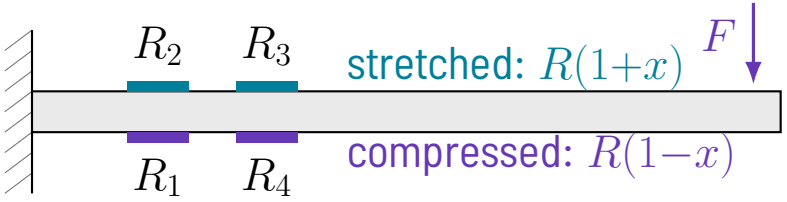
Quarter, half and full bridge



- quarter: $V_d \approx \frac{V_B}{4} x$ half: $V_d = \frac{V_B}{2} x$ full: $V_d = V_B\, x$ $(x = k\,\varepsilon)$
- half and full bridge are exactly linear: the leg current stays constant
- temperature changes all gauges equally: compensated

▶ Falstad: `examples/bridges/bridge_types.txt`

Where the gauges sit: bending beam



- strain $\varepsilon = \Delta l/l$, gauge factor $k \approx 2$: $x = \Delta R/R = k\,\varepsilon$
- top and bottom see the same strain with opposite sign
- stretched gauges into one diagonal (R_2, R_3), compressed into the other (R_1, R_4): the effects add up
- half bridge: R_1, R_2 only full bridge: all four

Example: quarter, half or full bridge?

- $R = 350\,\Omega$, $k = 2$, $\varepsilon = 1000\,\mu\text{m/m}$, $V_B = 5\,\text{V}$: $x = 0.002$, $\Delta R = 0.7\,\Omega$

	quarter	half	full
active gauges	1	2 (adjacent arms)	4
output V_d	$\approx \frac{V_B}{4}\,x$	$\frac{V_B}{2}\,x$	$V_B\,x$
example	2.50 mV	5.00 mV	10.0 mV
linearity	error $\approx -x/2$	exact	exact
temperature	dummy gauge needed	compensated	compensated
typical use	one sensor (Pt100)	bending beam	load cell, scale

- twice the active gauges, twice the signal: less gain, less amplified noise
- gain for 2 V: quarter 800, half 400, full 200

▶ Falstad: `examples/bridges/bridge_types.txt`

Exercise 5.2: Strain gauge bridge

Strain gauges with $R = 350\,\Omega$ and $k = 2$ are glued to a beam. The strain is $\varepsilon = 1000\,\mu\text{m/m}$, the bridge supply $V_B = 5\,\text{V}$.

a) Calculate ΔR.
b) Calculate the output of a quarter, a half and a full bridge.
c) Which gain is needed to get $2\,\text{V}$ at this strain from the full bridge?

Solution 5.2

a) $x = k\varepsilon = 2 \cdot 10^{-3}$, $\Delta R = 0.002 \cdot 350\,\Omega = 0.7\,\Omega$.

b) Quarter: $V_d = 5\,(350.7/700.7 - 0.5) = 2.4975$ mV ($\approx \frac{5}{4} \cdot 0.002 = 2.5$ mV); half: $\frac{5}{2} \cdot 0.002 = 5.00$ mV; full: $5 \cdot 0.002 = 10.0$ mV.

c) $G = 2\,\text{V}/10\,\text{mV} = 200$, with an instrumentation amplifier (chapter 6).

▶ Falstad: `examples/bridges/bridge_types.txt`:
the three voltmeters show 2.50 mV, 5.00 mV and 10.0 mV.

5.4 Resistive Temperature Sensors

Platinum sensors Pt100 and Pt1000

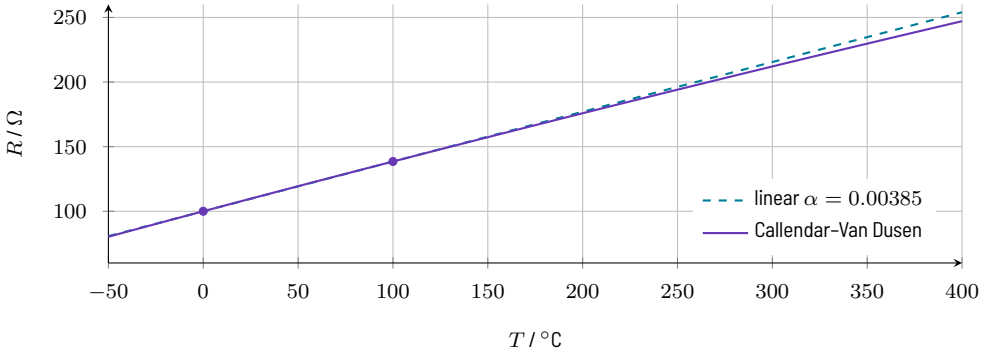
- linear model ($0\,°\text{C} \ldots 100\,°\text{C}$):

Pt100 / Pt1000

$$R(T) = R_0\,(1 + \alpha T), \quad R_0 = 100\,\Omega \,/\, 1000\,\Omega, \quad \alpha = 3.85 \times 10^{-3}/\text{K}$$

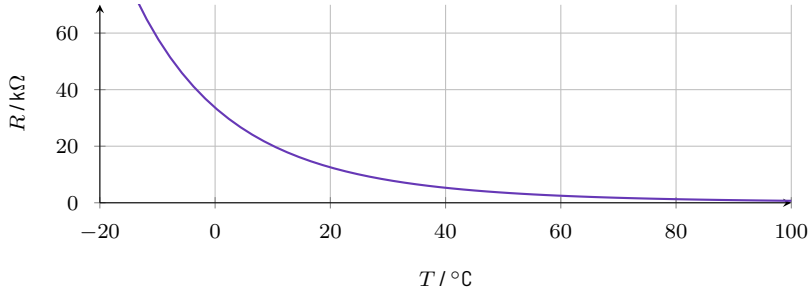
- Pt100: $100\,\Omega$ at $0\,°\text{C}$, $138.5\,\Omega$ at $100\,°\text{C}$, $0.385\,\Omega/\text{K}$
- Callendar–Van Dusen ($T \geq 0\,°\text{C}$): $R(T) = R_0(1 + AT + BT^2)$
  - $A = 3.9083 \times 10^{-3}/\text{K}$, $B = -5.775 \times 10^{-7}/\text{K}^2$
- tolerance class B: $\pm(0.3\,\text{K} + 0.005\,|T|)$, class A: $\pm(0.15\,\text{K} + 0.002\,|T|)$

Pt100 characteristic



- $0\,°C$ … $100\,°C$: linear model and standard agree at both ends, $0.4\,K$ apart at $50\,°C$
- $400\,°C$: $254\,\Omega$ (linear) vs. $247\,\Omega$: $18\,K$ error

NTC thermistors



- $R(T) = R_{25} \exp\left(B\left(\frac{1}{T} - \frac{1}{T_{25}}\right)\right)$, e.g. $R_{25} = 10\,\mathrm{k\Omega}$, $B = 3950\,\mathrm{K}$ (T in kelvin)
- sensitivity at $25\,°\mathrm{C}$: $-B/T^2 = -4.4\,\%/\mathrm{K}$ (Pt: $+0.39\,\%/\mathrm{K}$)
- strongly nonlinear: linearise with a series resistor, or a table in software

Self-heating: why the sensor current is limited

- the measuring current heats the sensor: $P = I^2 R$
- self-heating coefficient E: e.g. $0.1\,\mathrm{K/mW}$ (thin film, in water) … $1\,\mathrm{K/mW}$ (in still air)

Self-heating error

$$\Delta T = E \cdot I^2 R$$

- Pt100, 1 mA: $P = 0.1$ mW, $\Delta T = 0.01\,\mathrm{K} \ldots 0.1\,\mathrm{K}$
- Pt100, 10 mA: $P = 10$ mW, $\Delta T = 1\,\mathrm{K} \ldots 10\,\mathrm{K}$
- rule: Pt100 ≤ 1 mA, Pt1000 ≤ 0.3 mA (capstone: $I \leq 1$ mA)

Exercise 5.3: Pt100 values

a) Calculate the resistance of a Pt100 at $25\,^{\circ}\mathrm{C}$ and $37\,^{\circ}\mathrm{C}$ with the linear model.

b) Calculate $R(100\,^{\circ}\mathrm{C})$ and $R(200\,^{\circ}\mathrm{C})$ with the Callendar–Van Dusen equation. Which temperature does the linear model assign to these resistances?

c) A Pt100 is read with 5 mA in still air ($E = 0.5\,\mathrm{K/mW}$). How large is the self-heating error at $0\,^{\circ}\mathrm{C}$?

Solution 5.3

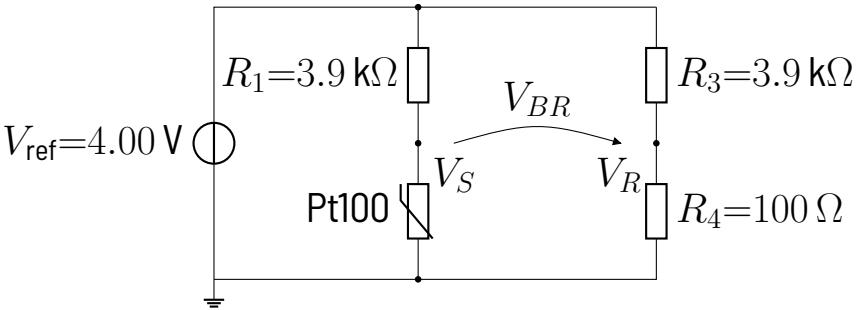
a) $R(25) = 100\,(1 + 0.00385 \cdot 25) = 109.63\,\Omega$, $R(37) = 114.25\,\Omega$.

b) $R(100) = 100\,(1 + 0.39083 - 0.005775) = 138.51\,\Omega$, linear model: $T = 0.3851/0.00385 = 100.0\,°\text{C}$. $R(200) = 100\,(1 + 0.78166 - 0.0231) = 175.86\,\Omega$ $\Rightarrow$ linear: $T = 197.0\,°\text{C}$ (3 K too low).

c) $P = (5\,\text{mA})^2 \cdot 100\,\Omega = 2.5\,\text{mW}$, $\Delta T = 1.25\,\text{K}$: four times the class B tolerance at $0\,°\text{C}$ ($0.3\,\text{K}$).

5.5 Designing the Sensor Bridge

The capstone bridge



- $R_4 = R_0 = 100\,\Omega$: balanced at $0\,°\text{C}$
- $I = V_{\text{ref}}/(R_1 + R) \leq 1\,\text{mA} \Rightarrow R_1 \geq 3.9\,\text{k}\Omega$; $R_3 = R_1$ $(m = 39)$
- $V_{BR}(T) = V_{\text{ref}} \left(\frac{R(T)}{R_1 + R(T)} - \frac{R_4}{R_3 + R_4} \right)$

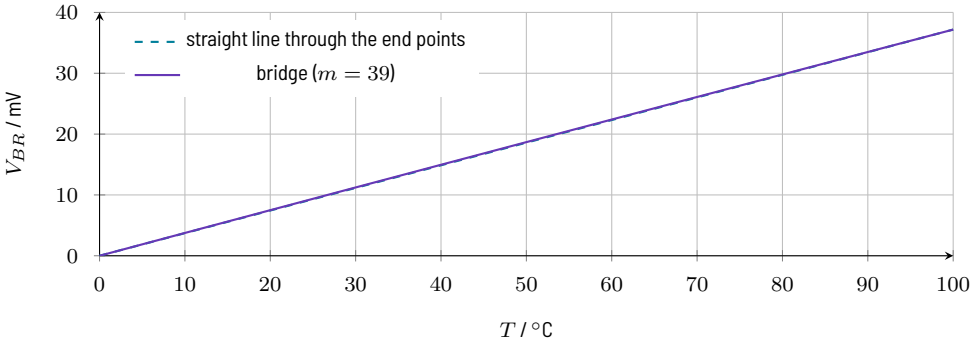
Numbers of the capstone bridge

	$0\,°C$	$50\,°C$	$100\,°C$
sensor R	$100\,\Omega$	$119.25\,\Omega$	$138.5\,\Omega$
sensor current	1.000 mA	0.995 mA	0.990 mA
V_S	100.0 mV	118.7 mV	137.2 mV
V_R	100.0 mV	100.0 mV	100.0 mV
V_{BR}	0 mV	18.68 mV	37.18 mV

- sensitivity ≈ 0.37 mV/K; nonlinearity at $50\,°C$: $+0.09$ mV ($0.25\,\%$ of full scale)
- required gain: $G = 4.00\,\text{V}/37.18\,\text{mV} = 107.6$
- self-heating: $P = 0.1$ mW

▶ Falstad: `examples/bridges/pt100_bridge.txt`

Bridge output over temperature



- with $m = 39$ the curve is almost straight: maximum deviation $0.09\,\text{mV} \approx 0.24\,\text{K}$
- with $m = 1$ (all arms $100\,\Omega$) the deviation would be $4.4\,\text{K}$

Design card: Sensor bridge

- excitation V_B: the ADC reference (ratiometric), here $4.00\,\text{V}$
- reference arm: $R_4 = R_{\text{sensor}}(T_{\text{min}}) \Rightarrow V_d = 0$ at T_{min}
- current limit: $R_1 \geq V_B / I_{\text{max}} - R_{\text{sensor}}$, round up to E-series; $R_3 = R_1$
- full-scale output $V_d(T_{\text{max}})$ with the exact formula, gain $G = V_{\text{FS}} / V_d(T_{\text{max}})$
- check nonlinearity ($\approx x/(m+1)$), self-heating ($I^2 R \cdot E$), common-mode voltage ($\approx V_B/(m+1)$)

Exercise 5.4: Design of the capstone bridge

A Pt100 ($\alpha = 3.85 \times 10^{-3}/\mathrm{K}$) shall measure $0\,°\mathrm{C} \ldots 100\,°\mathrm{C}$. The bridge is fed from $V_{\mathrm{ref}} = 4.00\,\mathrm{V}$, the sensor current must not exceed $1\,\mathrm{mA}$.

a) Choose R_1, R_3 and R_4.
b) Calculate the bridge output at $0\,°\mathrm{C}$, $50\,°\mathrm{C}$ and $100\,°\mathrm{C}$ and the sensor currents.
c) How large is the deviation from a straight line at $50\,°\mathrm{C}$, in millivolt and in kelvin?
d) Which total amplifier gain gives $4.00\,\mathrm{V}$ at $100\,°\mathrm{C}$?

Solution 5.4

a) $R_4 = 100\,\Omega$ (balance at $0\,°C$); $R_1 \geq 4\,V/1\,mA - 100\,\Omega = 3.9\,k\Omega$ (E12), $R_3 = R_1 = 3.9\,k\Omega$.

b) $V_R = 4 \cdot 100/4000 = 100.0\,mV$. $0\,°C$: $V_{BR} = 0$, $I = 1.000\,mA$. $50\,°C$: $V_S = 4 \cdot 119.25/4019.25 = 118.68\,mV$, $V_{BR} = 18.68\,mV$. $100\,°C$: $V_S = 4 \cdot 138.5/4038.5 = 137.18\,mV$, $V_{BR} = 37.18\,mV$, $I = 0.990\,mA$.

c) Straight line: $37.18/2 = 18.59\,mV$; deviation $0.09\,mV$, with $0.372\,mV/K$: $0.24\,K$.

d) $G = 4.00\,V/37.18\,mV = 107.6$ (capstone: 10×10.76).

▶ Falstad: `examples/bridges/pt100_bridge.txt`: with the slider at $100\,\Omega$ the voltmeter shows $0\,V$ and $V_R = 100\,mV$; at $119.25\,\Omega$ $18.68\,mV$; at $138.5\,\Omega$ $37.18\,mV$.

Exercise 5.5: Alternative: Pt1000 at 5 V

A Pt1000 is used in a bridge with $V_B = 5\,\text{V}$, $R_1 = R_3 = 4.7\,\text{k}\Omega$, $R_4 = 1\,\text{k}\Omega$.

a) Calculate the sensor current and the self-heating power at $0\,^\circ\text{C}$ and $100\,^\circ\text{C}$.

b) Calculate the bridge output at $100\,^\circ\text{C}$ and the required gain for $4.00\,\text{V}$.

c) How large is the deviation from the straight line at $50\,^\circ\text{C}$? Compare with the Pt100 design.

Solution 5.5

a) $I = 5/5700 = 0.877\,\text{mA}$, $P = I^2 \cdot 1\,\text{k}\Omega = 0.77\,\text{mW}$; at $100\,°\text{C}$: $I = 5/6085 = 0.822\,\text{mA}$, $P = 0.94\,\text{mW}$, almost ten times the Pt100.

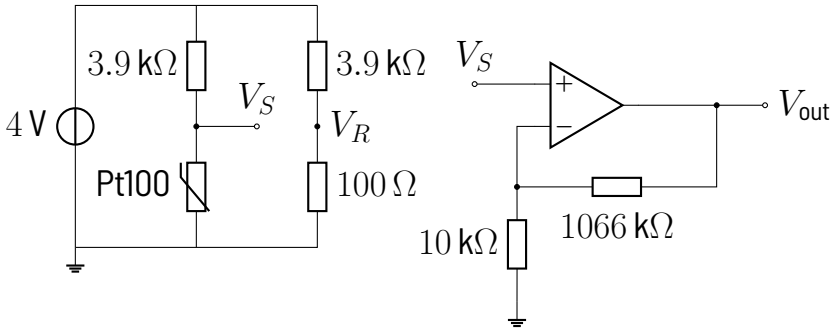
b) $V_d = 5\,(1385/6085 - 1000/5700) = 260.9\,\text{mV}$, $G = 4.00/0.2609 = 15.33$.

c) $V_d(50\,°\text{C}) = 5\,(1192.5/5892.5 - 0.17544) = 134.7\,\text{mV}$, straight line $130.4\,\text{mV}$: $4.3\,\text{mV} = 1.6\,\%$ of full scale ($m = 4.7$), almost seven times worse than the Pt100 design with $m = 39$; in exchange the signal is seven times larger.

▶ Falstad: `examples/bridges/pt1000_bridge.txt`: voltmeter $0\,\text{V}$, $134.7\,\text{mV}$ and $260.9\,\text{mV}$ at the slider positions 1000, 1192.5 and $1385\,\Omega$.

5.6 Reading the Bridge: Common-Mode Voltage

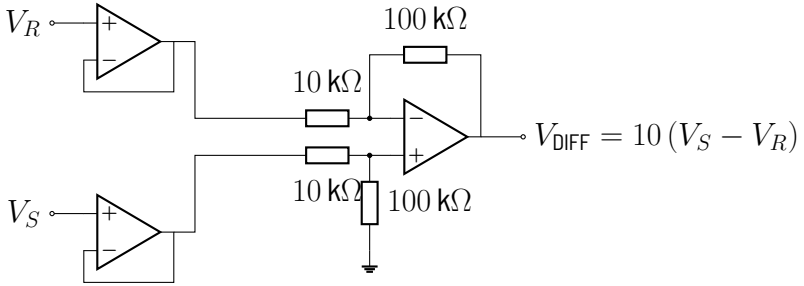
A single-ended amplifier fails



- gain 107.6 applied to $V_S = 100$ mV: $V_{\text{out}} = 10.75$ V at $0\,^\circ$C (wanted: 0 V)
- at $100\,^\circ$C: 14.76 V; the useful signal is a small change on a large offset
- any drift of V_{ref} appears amplified at the output

▶ Falstad: `examples/bridges/bridge_single_ended.txt`

Difference and instrumentation amplifier

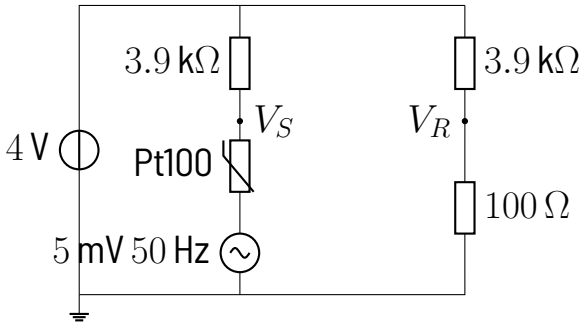


- difference amplifier: amplifies $V_S - V_R$, rejects V_{CM} (if the resistors match)
- buffers: no current from the bridge, the bridge is not loaded
- second stage $G_2 = 10.76$: $0\,\text{V}$ ... $4.00\,\text{V}$ (details: chapter 6)

▶ Falstad: `examples/bridges/bridge_ina.txt`

5.7 Interference and Lead Resistance

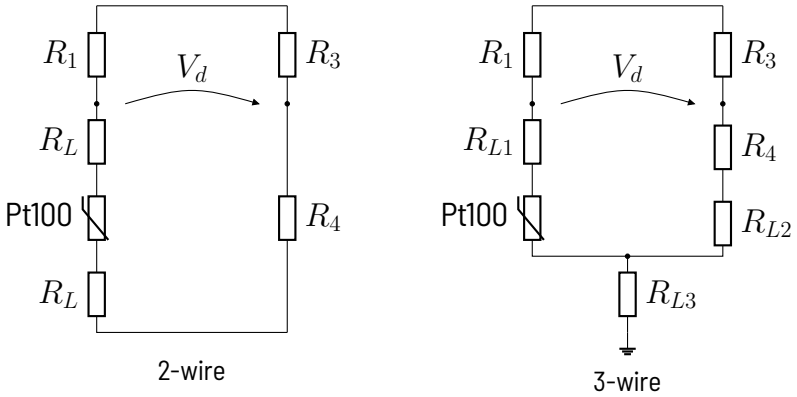
50 Hz in series with the sensor



- at V_S: $5\,\text{mV} \cdot \frac{R_1}{R_1+R} = 5\,\text{mV} \cdot \frac{3900}{4000} = 4.88\,\text{mV}$ amplitude
- after $G = 107.6$: $0.52\,\text{V}$ amplitude: about one LSB ($0.5\,\text{V}$)
- remedies: twisted pair, shield, short leads, low-pass filter before the ADC (chapters 6, 7)

▶ Falstad: `examples/bridges/bridge_50hz.txt`

Lead resistance: 2-wire and 3-wire connection



- 2-wire: $2R_L$ in series with the sensor; $R_L = 0.5\,\Omega$: $1\,\Omega \mathrel{\hat{=}} 2.6\,\mathrm{K}$
- 3-wire: R_{L1} in the sensor arm, R_{L2} in the neighbouring arm R_4: they cancel at balance; R_{L3} only carries the supply current
- 4-wire: current source through two wires, voltage sensed by the other two (precision instruments)

▶ Falstad: `examples/bridges/bridge_3wire.txt`

Exercise 5.6: Interference and lead resistance

The capstone bridge ($V_{\text{ref}} = 4\,\text{V}$, $R_1 = R_3 = 3.9\,\text{k}\Omega$, $R_4 = 100\,\Omega$, Pt100) is followed by an amplifier with gain 107.6.

a) A $5\,\text{mV}$, $50\,\text{Hz}$ interference voltage is induced in series with the sensor. Calculate its amplitude at V_S and at the amplifier output. Why does the instrumentation amplifier not reject it?

b) The sensor is connected with two leads of $0.5\,\Omega$ each. Which bridge output and temperature error result at $0\,^\circ\text{C}$?

c) Repeat b) for the 3-wire connection, at $0\,^\circ\text{C}$ and $100\,^\circ\text{C}$.

Solution 5.6

a) The source sees R_1 above and the sensor below: $5\,\text{mV} \cdot 3900/4000 = 4.875\,\text{mV}$ at V_S, $4.875\,\text{mV} \cdot 107.6 = 0.52\,\text{V}$ at the output. It changes only V_S, not V_R: it is a differential signal.

b) $R = 101\,\Omega$: $V_d = 4\,(101/4001 - 0.025) = 0.975\,\text{mV}$; with $0.372\,\text{mV/K}$: $2.6\,\text{K}$.

c) $0\,°\text{C}$: both lower arms have $100.5\,\Omega$, $V_d = 0$. $100\,°\text{C}$: $V_d = 37.16\,\text{mV}$ instead of $37.18\,\text{mV}$ (2-wire: $38.13\,\text{mV}$); the remaining error is $0.05\,\text{K}$.

▶ Falstad: `examples/bridges/bridge_50hz.txt`: the scope shows $4.88\,\text{mV}$ amplitude at V_S and $0.52\,\text{V}$ at `Vout`;

▶ Falstad: `examples/bridges/bridge_3wire.txt`: voltmeters $0.975\,\text{mV}$ (2-wire) and $0\,\text{V}$ (3-wire) at $0\,°\text{C}$, $38.13\,\text{mV}$ and $37.16\,\text{mV}$ with both sliders at $138.5\,\Omega$.